



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS (EXTENSION 2)

2015 HSC Course Assessment Task 1

Friday Dec 5, 2014

General instructions

- Working time – 60 minutes.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets + answer sheet used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer sheet provided (numbered as page 3)

SECTION II

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: **# BOOKLETS USED:**

Class (please ✓)

☐ 12M4A – Mr Lam

☐ 12M4B – Mr Ireland

☐ 12M4C – Mr Lin

Marker's use only.

QUESTION	1-5	6	7	8	Total	%
MARKS	$\overline{5}$	$\overline{14}$	$\overline{11}$	$\overline{15}$	$\overline{45}$	$\overline{100}$

Section I

5 marks

Attempt Question 1 to 5

Mark your answers on the answer sheet provided.

Questions

Marks

1. If $z = \sqrt{2} \left(\cos \left(-\frac{4\pi}{5} \right) + i \sin \left(-\frac{4\pi}{5} \right) \right)$ then z^9 is equal to: 1
- (A) $z = 16\sqrt{2} \left(\cos \left(\frac{36\pi}{5} \right) + i \sin \left(\frac{36\pi}{5} \right) \right)$
- (B) $z = 16\sqrt{2} \left(\cos \left(-\frac{\pi}{5} \right) + i \sin \left(-\frac{\pi}{5} \right) \right)$
- (C) $z = 16\sqrt{2} \left(\cos \left(\frac{4\pi}{5} \right) + i \sin \left(\frac{4\pi}{5} \right) \right)$
- (D) $z = 9\sqrt{2} \left(\cos \left(-\frac{\pi}{5} \right) + i \sin \left(-\frac{\pi}{5} \right) \right)$
2. If $P(z) = z^3 - 2z^2 + 4z - 8, z \in \mathbb{C}$, then a linear factor of $P(z)$ is 1
- (A) 2
- (B) $2i$
- (C) $z + 2$
- (D) $z + 2i$
3. The distance between the two points z and $-\bar{z}$ in the complex plane is given by 1
- (A) $2 \operatorname{Re}(z)$
- (B) $2 \operatorname{Im}(z)$
- (C) $|z|$
- (D) $2 \operatorname{Re}(z) + 2 \operatorname{Im}(z)$

4. $P(z)$ is a polynomial in z of degree 4 with real coefficients. Which of the following statements **must** be **false**. **1**
- (A) $P(z) = 0$ has no real roots.
- (B) $P(z) = 0$ has one real root and three non-real roots.
- (C) $P(z) = 0$ has two real roots and two non-real roots
- (D) $P(z) = 0$ has four real roots
5. Given that $(1 + i)^n = ai$, where a is a non-zero real constant, then $(1 + i)^{2n+2}$ simplifies to **1**
- (A) a^4
- (B) $2a^2i$
- (C) $1 + a^2i$
- (D) $-2a^2i$

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g. “●”

- | | | | | |
|-----|-----|-----|-----|-----|
| 1 – | (A) | (B) | (C) | (D) |
| 2 – | (A) | (B) | (C) | (D) |
| 3 – | (A) | (B) | (C) | (D) |
| 4 – | (A) | (B) | (C) | (D) |
| 5 – | (A) | (B) | (C) | (D) |

Examination continues overleaf...

Section II

40 marks

Attempt Questions 6 to 8

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (14 Marks)		Commence a NEW page.	Marks
(a)	Express $(2 - 3i)^2$ in the form $a + ib$, where a and b are real		2
(b)	z is the complex number $1 + i$ Write the following in modulus-argument form		
	i. z		2
	ii. iz		1
(c)	Sketch on the Argand diagram the locus $ z - 1 = z + 1 - 2i $		2
(d)	On an Argand diagram, graph the intersection of the regions defined by		4
	$z\bar{z} \geq 9, z + \bar{z} \leq 8$ and $0 < \arg z < \frac{\pi}{4}$		
(e)	Let ω be one of the non-real cube roots of 1		
	i. Show that $1 + \omega + \omega^2 = 0$		1
	ii. Hence or otherwise, prove that $(1 + \omega - \omega^2)^3 - (1 - \omega + \omega^2)^3 = 0$.		2

Question 7 (11 Marks)

Commence a NEW page.

Marks

- (a) Find constants A , B , and C so that **2**

$$\frac{1}{1+x+x^2+x^3} = \frac{A}{1+x} + \frac{Bx+C}{1+x^2}$$

- (b) Find the integers m and n such that $(x+1)^2$ is a factor of the polynomial **3**

$$P(x) = x^5 + 2x^2 + mx + n$$

- (c) If α , β , γ are the roots of $x^3 + mx + n = 0$, form the equation whose roots are **3**

$$\frac{1}{\alpha+\beta}, \frac{1}{\beta+\gamma}, \frac{1}{\gamma+\alpha}$$

- (d) The equation $x^3 + px^2 + qx + r = 0$ has one root equal to the sum of the other two. Show that $p^3 - 4pq + 8r = 0$ **3**

Examination continues overleaf...

Question 8 (15 Marks) Commence a NEW page. **Marks**

- (a) The four complex numbers z_1, z_2, z_3, z_4 are represented on the complex plane by the points A, B, C, D respectively. **4**

Given that

$$z_1 - z_2 + z_3 - z_4 = 0 \quad \text{and} \quad z_1 - iz_2 - z_3 + iz_4 = 0$$

use vectors to determine the possible shape(s) for the quadrilateral $ABCD$. Show all reasoning.

- (b) Consider the polynomial equation: $z^5 - i = 0$
- i. Find all the roots of $z^5 - i = 0$. You may leave the roots in the form of $\text{cis } \theta$. **2**
- ii. Hence show that **3**

$$(z - i) \left[z^2 - \left(2i \sin \frac{\pi}{10} \right) z - 1 \right] \left[z^2 + \left(2i \sin \frac{3\pi}{10} \right) z - 1 \right] = 0$$

- iii. Hence or otherwise deduce that $\sin \frac{3\pi}{10} - \sin \frac{\pi}{10} = \frac{1}{2}$ **2**
- (c) If $\frac{z - 1 + i}{z + 1 - i} = ki$, where $k \in \mathbb{R}$ **4**

By drawing a diagram or otherwise, find the value of $|z|$. Show all reasoning.

End of paper.

Multiple Choice

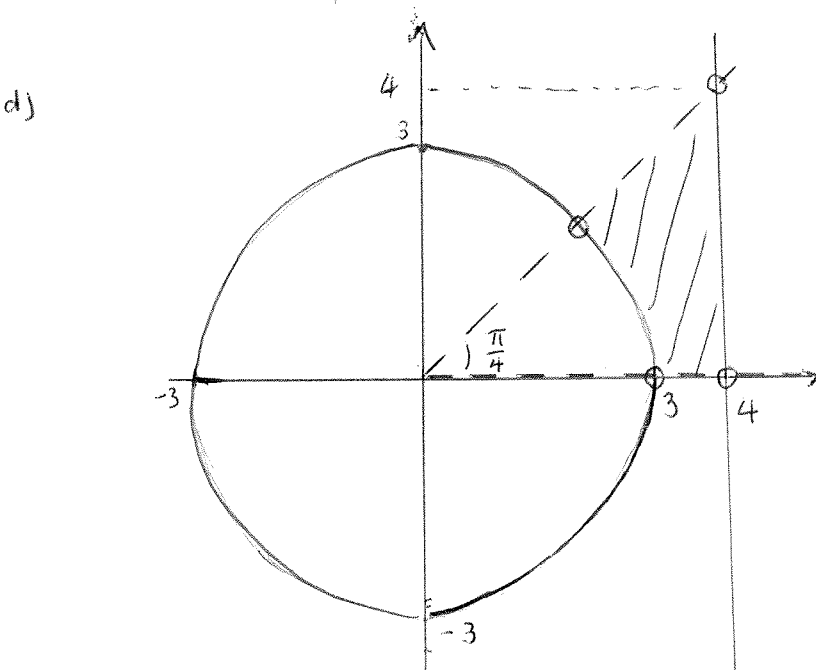
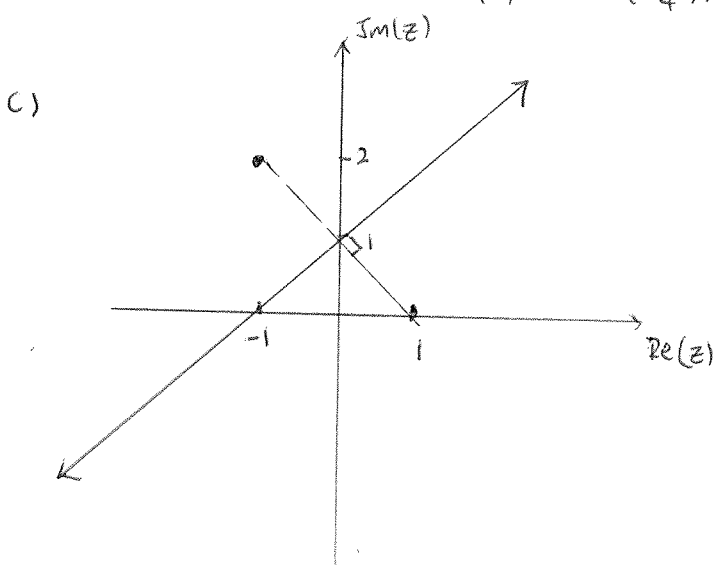
1. C 2. D 3. A 4. B 5. D

Question 6

a) $(2-3i)^2 = 4 + 2(-3i) \times 2 + 9i^2$
 $= -5 - 12i$

b) (i) $1+i = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$

(ii) $iZ = \sqrt{2} \left(\cos \left(\frac{\pi}{4} + \frac{\pi}{2} \right) + i \sin \left(\frac{\pi}{4} + \frac{\pi}{2} \right) \right)$
 $= \sqrt{2} \left(\cos \left(\frac{3\pi}{4} \right) + i \sin \left(\frac{3\pi}{4} \right) \right)$



e) i) Since ω is a non real cube root of $z^3 - 1 = 0$
 and $(z-1)(1+z+z^2) = 0$ upon factorising

then $1+\omega+\omega^2 = 0$ as $\omega - 1 \neq 0$

ii) As $1+\omega+\omega^2 = 0$
 $\Rightarrow 1+\omega = -\omega^2$
 and $1+\omega^2 = -\omega$

$$\begin{aligned} \text{LHS} &= (1+\omega-\omega^2)^3 - (1-\omega+\omega^2)^3 \\ &= (-\omega^2-\omega^2)^3 - (-\omega-\omega)^3 \\ &= (-2\omega^2)^3 - (-2\omega)^3 \\ &= -8(\omega^3)^2 - (-8\omega^3) \\ &= -8 + 8 \quad (\text{as } \omega^3 = 1) \\ &= 0 \\ &= \text{RHS} \end{aligned}$$

$$z\bar{z} \geq 9 \Rightarrow x^2 + y^2 \geq 9$$

$$\begin{aligned} z + \bar{z} &\leq 8 \Rightarrow 2\text{Re}(z) \leq 8 \\ \text{Re}(z) &\leq 4 \\ x &\leq 4 \end{aligned}$$

Question 7

$$a) \frac{1}{1+x+x^2+x^3} \equiv \frac{A}{1+x} + \frac{Bx+C}{1+x^2}$$

$$\Rightarrow A(1+x^2) + (1+x)(Bx+C) \equiv 1$$

$$\text{when } x = -1$$

$$2A = 1$$

$$A = \frac{1}{2}$$

$$\text{when } x = 0$$

$$A + C = 1$$

$$C = \frac{1}{2}$$

equating coefficient of x^2

$$A + B = 0$$

$$B = -\frac{1}{2}$$

$$b) P'(x) = 5x^4 + 4x + m$$

Because there is a double root
at $x = -1$

$$\therefore P'(-1) = 0$$

$$P'(-1) = 5 - 4 + m = 0$$

$$\therefore m = -1$$

$$\text{But } P(-1) = 0$$

$$\therefore P(-1) = (-1)^5 + 2(-1)^2 - (-1) + n = 0$$

$$\therefore 2 + n = 0$$

$$n = -2$$

$$c) x^3 + mx + n = 0$$

Sum of the roots

$$\alpha + \beta + \gamma = 0$$

$$\therefore \alpha + \beta = -\gamma$$

$$\therefore \frac{1}{\alpha + \beta} = -\frac{1}{\gamma}$$

Similarly

$$\frac{1}{\beta + \gamma} = -\frac{1}{\alpha}$$

$$\text{and } \frac{1}{\alpha + \gamma} = -\frac{1}{\beta}$$

Substitute x with $-\frac{1}{x}$

$$\left(-\frac{1}{x}\right)^3 + m\left(-\frac{1}{x}\right) + n = 0$$

$$\therefore nx^3 - mx^2 - 1 = 0$$

$$d) x^3 + px^2 + qx + r = 0$$

roots are α, β, γ

$$\therefore \alpha = \beta + \gamma$$

sum of the roots: $2\beta + 2\gamma = -p$

$$\therefore \beta + \gamma = -\frac{p}{2}$$

Now $\beta + \gamma$ is a root $\therefore -\frac{p}{2}$ is a root

Sub $-\frac{p}{2}$ in $x^3 + px^2 + qx + r = 0$

$$\left(-\frac{p}{2}\right)^3 + p\left(-\frac{p}{2}\right)^2 + q\left(-\frac{p}{2}\right) + r = 0$$

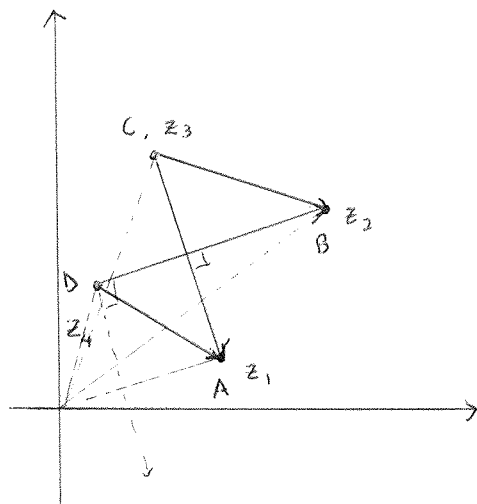
$$-\frac{p^3}{8} + \frac{p^3}{4} - \frac{pq}{2} + r = 0$$

multiply through by 8

$$p^3 - 4pq + 8r = 0$$

Question 8

a)



$$\text{If } z_1 - z_2 + z_3 - z_4 = 0$$

$$z_1 - z_4 = z_2 - z_3$$

$$\vec{CB} = \vec{DA}$$

$$\Rightarrow CB = AD$$

$$CB \parallel AD$$

$\therefore$ ABCD is a parallelogram

(One pair of opposite sides equal in length and parallel)

$$\text{Now if } z_1 - iz_2 - z_3 + iz_4 = 0$$

$$z_1 - z_3 = i(z_2 - z_4)$$

$\Rightarrow$ AC is a image of BD with rotation of 90°

i.e. $AC \perp BD$ and $AC = BD$

$\therefore$ diagonals are perpendicular and equal in length

$\therefore$ ABCD can only be a square.

b) i) $z^5 - i = 0 \Rightarrow z^5 = i = \text{cis}\left(\frac{\pi}{2} + 2k\pi\right), k \in \mathbb{Z}$

$$z = i^{\frac{1}{5}} = \left(\text{cis}\left(\frac{\pi}{2} + 2k\pi\right)\right)^{\frac{1}{5}}$$

$\therefore$ using de Moivre's Theorem

$$z = \text{cis}\left(\frac{\pi + 4k\pi}{10}\right)$$

For 5 roots, $k = -2, -1, 0, 1, 2$

$$z_1 = \text{cis}\left(-\frac{7\pi}{10}\right) \quad z_4 = \text{cis}\left(\frac{5\pi}{10}\right) = i$$

$$z_2 = \text{cis}\left(-\frac{3\pi}{10}\right) \quad z_5 = \text{cis}\left(\frac{9\pi}{10}\right)$$

$$z_3 = \text{cis}\left(\frac{\pi}{10}\right)$$

ii) Now $z^5 - i = (z - i)(z - \text{cis}\frac{\pi}{10})(z - \text{cis}\frac{9\pi}{10})(z - \text{cis}(-\frac{3\pi}{10}))(z - \text{cis}(-\frac{7\pi}{10}))$

For $(z - \text{cis}\frac{\pi}{10})(z - \text{cis}\frac{9\pi}{10})$

$$= z^2 - z \text{cis}\frac{9\pi}{10} - z \text{cis}\frac{\pi}{10} + \text{cis}\frac{\pi}{10} \text{cis}\frac{9\pi}{10}$$

$$= z^2 - z \left[\cos\frac{9\pi}{10} + i\sin\frac{9\pi}{10} + \cos\frac{\pi}{10} + i\sin\frac{\pi}{10} \right] + \text{cis}\pi$$

$$= z^2 - z \left[-\cos\left(\frac{\pi}{10}\right) + i\sin\frac{\pi}{10} + \cos\frac{\pi}{10} + i\sin\frac{\pi}{10} \right] - 1$$

$$\text{as } \cos(\pi - x) = -\cos x$$

$$\text{and } \sin(\pi - x) = \sin x$$

$$\cos \pi + i\sin \pi = -1$$

$$= z^2 - z \left[2i\sin\frac{\pi}{10} \right] - 1$$

$$\text{similarly for } (z - \text{cis}(-\frac{3\pi}{10}))(z - \text{cis}(-\frac{7\pi}{10}))$$

$$= z^2 - z \left[\text{cis}(-\frac{3\pi}{10}) + \text{cis}(-\frac{7\pi}{10}) \right] + \text{cis}(-\frac{3\pi}{10})\text{cis}(-\frac{7\pi}{10})$$

$$= z^2 - z \left[\cos(-\frac{3\pi}{10}) + i\sin(-\frac{3\pi}{10}) + \cos(-\frac{7\pi}{10}) + i\sin(-\frac{7\pi}{10}) \right] + \text{cis}(-\pi)$$

$$= z^2 - z \left[\cos\frac{3\pi}{10} - i\sin\frac{3\pi}{10} + \cos\frac{7\pi}{10} - i\sin\frac{7\pi}{10} \right] - 1 \quad \text{as } \cos \theta \text{ is even}$$

$\sin \theta$ is odd

$$= z^2 - z \left[\cos\frac{3\pi}{10} - i\sin\frac{3\pi}{10} - \cos\frac{3\pi}{10} - i\sin\frac{3\pi}{10} \right] - 1$$

$$= z^2 + z \left[2i\sin\frac{3\pi}{10} \right] - 1$$

$$\therefore z^5 - i = (z - i)(z^2 - (2i\sin\frac{\pi}{10})z - 1)(z^2 + (2i\sin\frac{3\pi}{10})z - 1) = 0$$

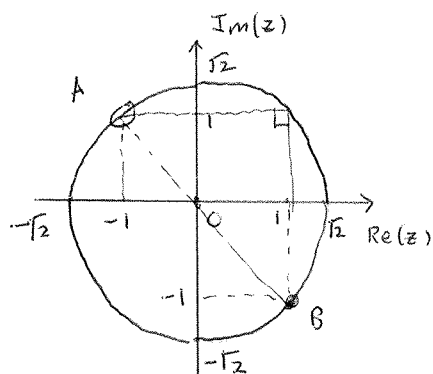
ii) From sum of the roots

$$i + \text{cis}\frac{\pi}{10} + \text{cis}\frac{9\pi}{10} + \text{cis}(-\frac{3\pi}{10}) + \text{cis}(-\frac{7\pi}{10}) = 0$$

$$\text{From (ii)} \quad 2i\sin\frac{\pi}{10} - 2i\sin\frac{3\pi}{10} = -i$$

$$\sin\frac{3\pi}{10} - \sin\frac{\pi}{10} = \frac{1}{2}$$

$$c) \arg\left(\frac{z - (1-i)}{z - (-1+i)}\right) = \pm \frac{\pi}{2}$$



$\therefore$ circle with origin $(0,0)$
radius $\sqrt{2}$

$$\therefore |z| = \sqrt{2}$$

Note that $(1, -1)$ is

actually part of the solution

- Forms a right angle at circumference

$\therefore$ AB is the diameter of a circle